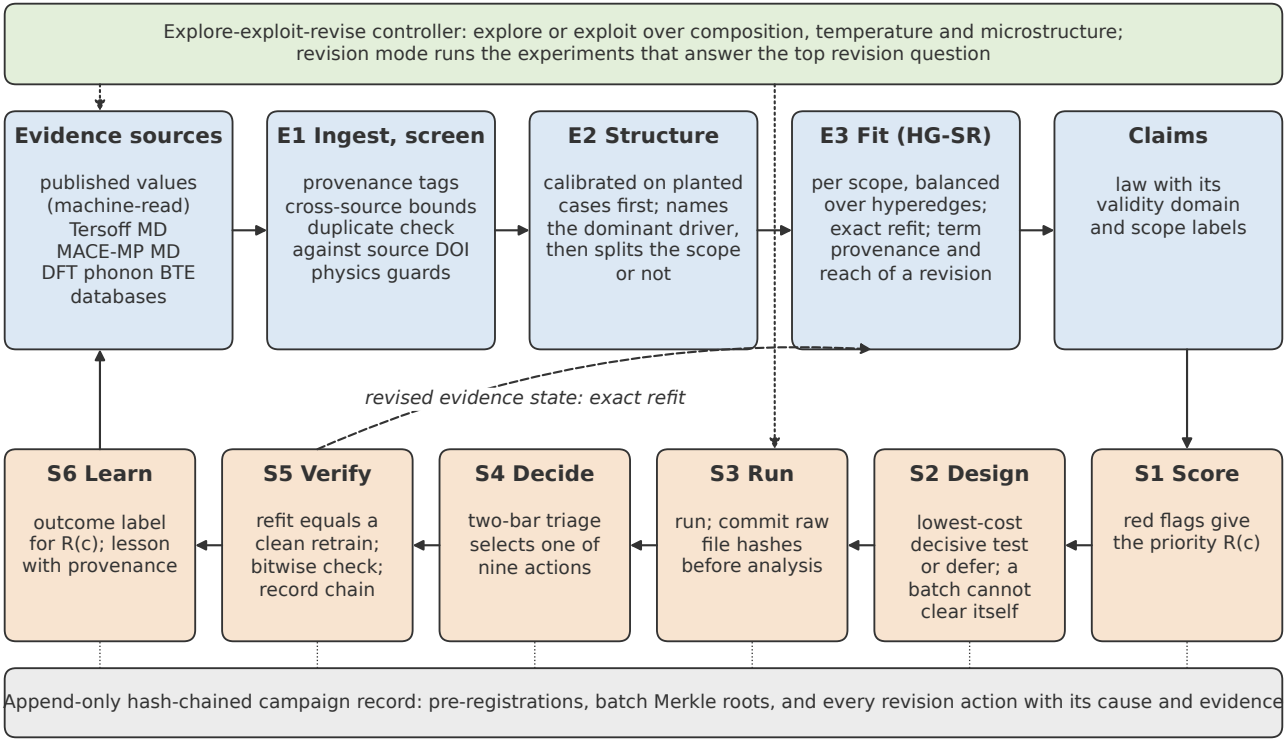
Explore-exploit-revise controller: explore or exploit over composition, temperature and microstructure;
revision mode runs the experiments that answer the top revision question
Evidence sources
published values
(machine-read)
Tersoff MD
MACE-MP MD
DFT phonon BTE
databases
E1 Ingest, screen
provenance tags
cross-source bounds
duplicate check
against source DOI
physics guards
E2 Structure
calibrated on planted
cases first; names
the dominant driver,
then splits the scope
or not
E3 Fit (HG-SR)
per scope, balanced
over hyperedges;
exact refit; term
provenance and
reach of a revision
Claims
law with its
validity domain
and scope labels
revised evidence state: exact refit
S6 Learn
outcome label
for R(c); lesson
with provenance
S5 Verify
refit equals a
clean retrain;
bitwise check;
record chain
S4 Decide
two-bar triage
selects one of
nine actions
S3 Run
run; commit raw
file hashes
before analysis
S2 Design
lowest-cost
decisive test
or defer; a
batch cannot
clear itself
S1 Score
red flags give
the priority R(c)
Append-only hash-chained campaign record: pre-registrations, batch Merkle roots, and every revision action with its cause and evidence